Response to Reviewers

Dear Editor and Reviewers,

We thank you for the careful evaluation of our manuscript and for the constructive comments that have improved its clarity, rigor, and reproducibility. Below we provide a point-by-point summary of the revisions made. For clarity, reviewer comments are paraphrased in italics, followed by our responses.

Reviewer 1

**Comment 1. *The machine learning framework requires clearer justification and improved robustness, particularly regarding class imbalance and model stability.* **

Response: We substantially revised the Random Forest (RF) workflow to address these concerns:

- The number of trees was increased from 300 to **500** to improve model stability.
- Class imbalance was explicitly addressed by capping the number of training samples at **50,000 observations per class**.
- Importantly, this sampling is applied **after the train/test split**, ensuring that validation metrics reflect the original class distribution.
- Leave-one-out (LOO) validation models were updated to use the same configuration (500 trees; 50,000 samples per class), ensuring consistency between validation and final model outputs.

The Methods section has been revised to state:

"To reduce class imbalance while maintaining sufficient representation of dominant classes, the training dataset was balanced by randomly sampling up to 50,000 observations per class. Random forest models were trained using 500 trees."

**Comment 2. *The observed model artifacts suggest potential issues with model configuration or training data.* **

Response: We clarified that the previously observed artifacts were primarily due to **class imbalance**, not insufficient model complexity. The revised manuscript now emphasizes that training data composition is the dominant control on model behavior, whereas increasing the number of trees alone does not resolve bias.

Reviewer 2

Comment 3. *The unsupervised clustering approach requires clearer justification and methodological transparency. *

Response: The K-means clustering workflow was revised as follows:

- The number of clusters was standardized to **k = 6 for both PlanetScope and Landsat datasets**, based on combined evaluation using elbow and silhouette methods.
- The clustering function was improved to remove reliance on global variables and to ensure consistent handling of raster structure.
- NA handling was refined to operate on a **per-date basis**, preventing unnecessary loss of valid observations.

The manuscript now clearly describes K-means as a step to partition spectral space prior to supervised classification.

Comment 4. *Geospatial preprocessing and data provenance require additional clarification. *

Response: We expanded the preprocessing section to include:

- Explicit identification of data levels:
  - PlanetScope imagery as **surface reflectance (Level 3B)**
  - Landsat imagery as **Collection 2 Level-2 surface reflectance**
- Clarification of temporal alignment between datasets.
- Documentation of a second set of PlanetScope dates used for mesohabitat comparison.
- Improved description of raster stacking and band standardization procedures.

Reviewer 3

Comment 5. *The mesohabitat classification workflow requires clearer methodological justification and improved efficiency. *

Response: We revised the mesohabitat workflow to improve both interpretability and performance:

- Mesohabitat classes are now derived directly from **depth and velocity rasters** using a consistent classification function.
- Median conditions are computed from **median depth and velocity**, rather than from classified rasters.

- Raster processing was optimized to reduce redundant reprojection and resampling operations.
- A revised color scheme was implemented to improve interpretability, with pools represented by blue gradients and high-velocity environments by warm colors.

Reviewer 4

**Comment 6. *Statistical interpretation requires clarification, particularly regarding correlations between land cover classes.* **

Response: The statistical analysis section was revised to:

- Clearly distinguish between moderate negative correlations observed in semi-supervised models and weak relationships in supervised models.
- Provide explicit geomorphic interpretation linking these relationships to river dynamics (submergence vs exposure).
- Improve explanation of differences between PlanetScope and Landsat results in terms of spatial resolution and sensitivity to fine-scale features.

Reviewer 5

**Comment 7. *Figures require improvement in clarity, layout, and interpretability.* **

Response: All figures were redesigned to improve visual quality and communication:

- The study area figure was completely restructured using a hierarchical layout with a dominant study-reach panel and supporting locator maps.
- Time series figures were consolidated to reduce redundancy and improve readability.
- Model comparison figures were refined with improved legend placement, consistent panel sizing, and reduced clutter.

Reviewer 6

**Comment 8. *Tables and overall manuscript organization require simplification and improved clarity.* **

Response:

- Tables were reorganized to reduce redundancy and improve readability.
- Average and total values were grouped logically.
- Formatting issues (spacing, alignment, and LaTeX rendering) were corrected.

Additional Revisions

Beyond reviewer comments, we implemented the following improvements:

- Enhanced methodological clarity throughout the manuscript.
- Improved consistency in terminology and notation.
- Refined figure referencing (use of “Fig.”) and LaTeX formatting.
- Removed redundant or overly interpretive language.

Summary

The revised manuscript includes:

- Improved model robustness through balanced training data
- Greater methodological transparency and reproducibility
- Clearer linkage between geomorphology and model outputs
- Substantially improved figures and tables

We believe these revisions fully address the reviewer comments and significantly strengthen the manuscript.

Thank you again for your time and consideration.

Sincerely,
The Authors